

Labid Al Nahiyen

Software Engineer

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TECHNICAL SKILLS

Languages	Python, C/C++, Java, TypeScript/JavaScript, SQL
Backend & APIs	REST API design, Express.js, Zustand
Frontend	React.js, SvelteKit, Tailwind CSS
Data & Infra	PostgreSQL, Docker, Git, Linux, Hugging Face
ML/AI (applied)	PyTorch, Distributed training (DDP), LangChain, LangGraph, RAG systems

WORK EXPERIENCE

TigerIT Bangladesh Ltd.

Aug 2025 – Present

Machine Learning Engineer (R&D)

- Built and maintained a cross-language production pipeline for a biometric matching system.
- Diagnosed and fixed compounding bugs across the native/managed boundary, including coordinate-transform and rotation-angle errors in the C++ DLL and image-canvas artifacts, shipped corrected DLL builds with reproducible build scripts.
- Engineered training infrastructure in PyTorch (multi-GPU distributed training) and resolved production-blocking issues: memory-scaling failures on an 800K+ class output layer, gradient-computation bugs from duplicated forward passes, and host out-of-memory crashes.
- Reduced FNMR by **41%** ($0.0151 \rightarrow 0.0089$ @ $\text{FMR} \leq 10^{-2}$) on the **NIST MINEX III** benchmark by optimizing the minutiae-based template generator and matching pipeline.
- Built internal debugging/visualization tools (patch inspection, rotation comparison, angle diagnostics) used by the team to isolate pipeline bugs faster.

Technologies: Python, C++, C#, CUDA, PyTorch, OpenCV, Git

Intellesphere.AI

May 2025 – Jul 2025

Full-Stack Developer

- Built responsive frontend components from UI/UX designs and integrated RESTful APIs for dynamic data rendering in a live production application.
- Tested, debugged, and optimized the deployed web application to ensure full functionality and stability for end users.

Technologies: React.js, Express.js, Zustand

SELECTED PROJECTS

Shop Genie - AI E-commerce Assistant

[GitHub](#)

SvelteKit, LLM, RAG

Full-stack shopping assistant enabling natural-language product search over a corpus of 10,000+ scraped listings; built the retrieval pipeline, backend service layer, and SvelteKit frontend end-to-end.

NerdHerd - Educational Platform

[GitHub](#)

SvelteKit, Tailwind CSS, PostgreSQL

All-in-one student platform with live classes, video calls, community forums, and quiz modules; designed the PostgreSQL schema and built core application features.

Subset-of-C Compiler

[GitHub](#)

C, Bison, Flex, YACC

Implemented lexical analysis, parsing, symbol table management, and assembly code generation for a C-language subset, full compiler front-to-back-end pipeline.

Bidirectional Autonomous Robot

[Website](#)

Arduino, Embedded C, Hierarchical Q-Learning

Autonomous crawling robot with real-time obstacle avoidance and live policy adaptation on a custom embedded Arduino setup.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Feb 2020 – Mar 2025

B.Sc. in Computer Science and Engineering

Notable Courses: Data Structures & Algorithms, Software Engineering, Database Systems, Compiler Design, OOP, Computer Security, Machine Learning

RESEARCH & PUBLICATIONS

When Safety Blocks Sense: Measuring Semantic Confusion in LLM Refusals

2025

Co-Author | Preprint: [arXiv:2512.01037](#)

Contextually Appropriate Loss Function for Dense Crowd Tracking

2024

Undergraduate Thesis | Supervisor: Prof. A. B. M. Alim Al Islam, BUET

HONORS & AWARDS

Champion – OpenAPI Hackathon, 11th National ICT Fest

2024

IUT, Bangladesh

7th Place – SEC Inter-University Junior Programming Contest

2022

Team: BUET Fast & Furious

REFERENCES

Prof. A. B. M. Alim Al Islam

Professor, Computer Science and Engineering

Bangladesh University of Engineering and Technology

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